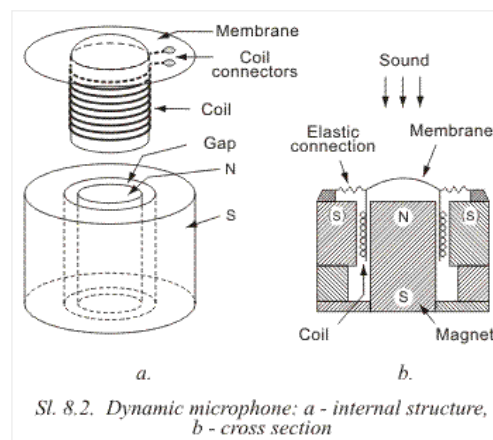


8.1 MICROPHONES

MIKROELEKTRONIKA

There are several different types of microphone: carbon, dynamic, crystal, capacitive (electret). Carbon microphones were one of the first to be invented and were used mainly in telephone applications. But they are very noisy as the carbon granules rattle when the microphone is moved and this type is being replaced by more advanced types.



Dynamic microphones are in wide use and their quality of reproduction is superb. They are used in the recording industry for music and speech where high fidelity is required. Basically they are exactly the same as a speaker, the only difference being the size. But their only limitation is the very low output. The internal structure is shown in figure 8.2. A paper cylinder, onto which fine copper wire is wound, is connected to a membrane which moves under the force of sound pressure created by the sound source. This coil is in a narrow gap with a high magnetic field created by a permanent magnet. When the coil moves in this magnetic field, it produces a voltage identical to the sound causing the movement.

Because of the low resistance (impedance) of a dynamic microphone, it usually needs a transformer so it can be connected to an amplifier (called a pre-amp). This transformer is usually built into the microphone's case, but if it is absent, it is necessary to connect the microphone to a preamplifier with low input resistance.

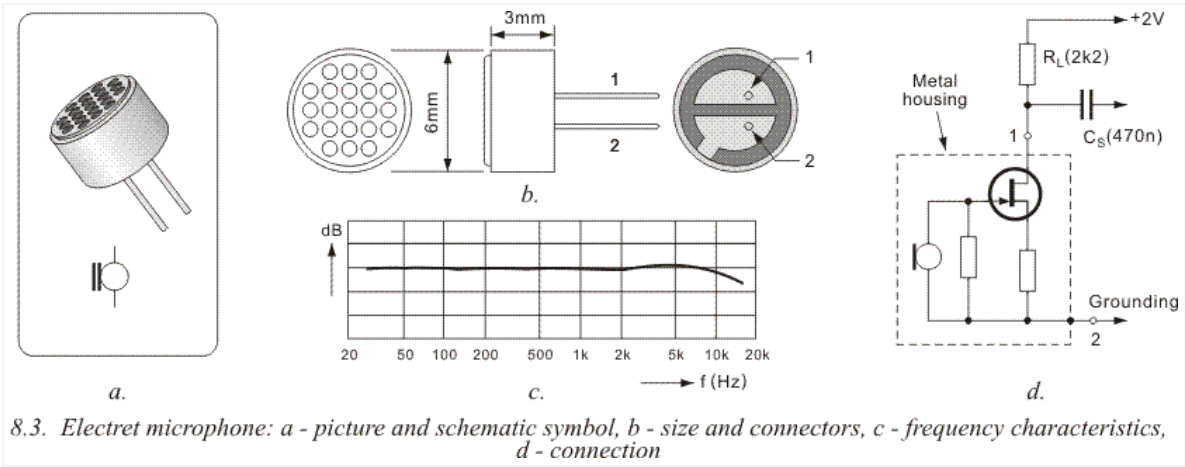
Crystal microphones contain a crystal called a "piezo crystal" that is connected to a small diaphragm. When sound waves hit the diaphragm, the crystal changes shape and it produces a voltage. This voltage is passed to an amplifier.

Recently, electret microphones have improved in quality and taken over from nearly all other types of microphone. They are small, rugged, low in price and produce a very high quality output.

The shape, size and characteristics are shown in 8.3.

The microphone contains a Field Effect Transistor, which means it needs a DC voltage for it to operate. Figure 8.3d shows how an electret mic is connected to a circuit. It needs a "load resistor" to limit the current to the FET and the output is taken across this resistor. That's the technical way of saying the output is taken from the point

where the resistor meets the FET.



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